

Cross-Platform Installation Guide

WSL2, Docker Desktop, and Ollama for Windows & Mac Users

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Abstract

This guide walks students through installing the three prerequisites needed for the Goose + GOWA local agent stack: **WSL2** (Windows only), **Docker Desktop** (both platforms), and **Ollama** (both platforms). Each section includes verification commands and platform-specific gotchas, particularly the classic WSL2 “permission denied” error that blocks first-time Docker users.

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Part I

Windows Installation

1 Prerequisites Check

Before starting, verify your Windows version and ensure virtualization is enabled.

- **Check Windows Version:** Open PowerShell and run `winver`. You need **Windows 11** or **Windows 10 Build 2004+** (Build 19041 or higher).
- **Enable Virtualization:** Open PowerShell as Administrator and run `systeminfo`. Look for the line:

```
Hyper-V Requirements:  Virtualization Enabled In Firmware:  Yes
```

If it says “No”, restart your PC, enter BIOS/UEFI, and enable **Intel VT-x** or **AMD-V**.

2 Install WSL2 and Ubuntu

The easiest method is a single PowerShell command.

1. **Open PowerShell as Administrator:** Press **Win + X** and select “Windows PowerShell (Admin)” or “Terminal (Admin)”.
2. **Install WSL:** Run the following command. This enables WSL, installs the Linux kernel, sets WSL2 as default, and installs Ubuntu (latest LTS).

```
wsl --install
```

3. **Reboot:** Restart your computer when prompted.
4. **Update WSL Kernel:** After reboot, open PowerShell again and run:

```
wsl --update
```

5. **Create User Account:** Launch “Ubuntu” from the Start Menu. It will prompt you to create a **username** (lowercase, no spaces) and **password**. Remember this password — you will need it for `sudo` commands.

3 Install Docker Desktop

Docker Desktop will use WSL2 as its backend for better performance.

1. **Download:** Get the installer from <https://www.docker.com/products/docker-desktop>.
2. **Install:** Run `Docker Desktop Installer.exe`. Keep the default settings and ensure “**Use WSL 2 instead of Hyper-V**” is checked.
3. **Launch:** After installation, search for and open **Docker Desktop**. Accept the service agreement.

4. **WSL Integration:** In Docker Desktop, go to **Settings > Resources > WSL Integration**. Ensure “**Enable integration with my default WSL distro**” is checked, and toggle on your Ubuntu distribution. This allows the `docker` command to work inside WSL2.
5. **Verify Installation:** Open your Ubuntu terminal and run:

```
docker --version
```

It should print the installed version (e.g., `Docker version 27.3.1, build ...`).

3.1 Critical Gotcha — Add Your User to the `docker` Group

By default, the `docker` command requires `sudo` because the Docker daemon socket is owned by `root`. If you try to run a Docker command without `sudo`, you will see an error like:

```
permission denied while trying to connect to the Docker daemon socket
at unix:///var/run/docker.sock
```

To fix this permanently, add your WSL2 user to the `docker` group:

```
# Add your user to the docker group
sudo usermod -aG docker $USER

# Verify the group was added
groups $USER
```

You Must Log Out and Back In

The group change does not take effect in the current shell. You must restart WSL for it to apply:

```
# Exit all WSL terminals, then in PowerShell:
wsl --shutdown
```

Reopen Ubuntu from the Start Menu, then verify:

```
docker ps
```

If it works without `sudo`, the group change succeeded.

3.2 Verify Docker is Working Correctly

Run these checks in order. If any fails, fix it before proceeding.

Check the Docker daemon is running:

```
docker info
```

Should print server details without errors. If you see `Cannot connect to the Docker daemon`, ensure Docker Desktop is running.

Run the hello-world container:

```
docker run hello-world
```

Expected output ends with `Hello from Docker!` — this confirms Docker can pull and run images.

Check Docker Compose:

```
docker compose version
```

Should print a version like `Docker Compose version v2.29.7`.

List running containers:

```
docker ps
```

Should show an empty table (headers only) — that is success. If you see permission errors, revisit the group gotcha above.

Check Docker disk usage (useful before pulling large images like GOWA):

```
docker system df
```

Shows space used by images, containers, and volumes.

3.3 Start Docker Desktop Automatically

To avoid manually launching Docker Desktop after every reboot:

- Open **Docker Desktop** → **Settings** → **General**
- Check “**Start Docker Desktop when you sign in to your computer**”

This ensures `docker` commands in WSL2 work immediately after a reboot.

4 Install Ollama

Ollama runs natively on Windows and uses your GPU (NVIDIA/AMD) if available.

1. **Quick Install:** Open PowerShell and run the official install script:

```
irm https://ollama.com/install.ps1 | iex
```

Alternatively, download `OllamaSetup.exe` from <https://ollama.com/download/windows> and run it.

2. **Protect C: Drive (Critical):** Models are large. Set a custom storage path *before* downloading models.
 - Press Win + R, type `sysdm.cpl`, and go to **Advanced > Environment Variables**.
 - Under **User variables**, click **New**.
 - Variable name: `OLLAMA_MODELS`
 - Variable value: `D:\OllamaModels` (or your preferred path).

3. **Verify:** Open a new PowerShell window and run:

```
ollama --version
```

Part II

Mac Installation

5 Prerequisites Check

- **System:** macOS 14 (Sonoma) or newer is recommended for Docker Desktop.
- **Hardware:** At least **16 GB of RAM** is recommended for running local LLMs efficiently. Apple Silicon (M1/M2/M3) chips offer significantly better performance than Intel Macs.

6 Install Docker Desktop

1. **Download:** Choose the correct version for your Mac from <https://www.docker.com/products/docker-desktop>:
 - **Apple Silicon:** For M1, M2, M3, etc. chips.
 - **Intel Chip:** For older Macs.
2. **Install:** Double-click the downloaded `.dmg` file, then drag the **Docker icon** into your **Applications** folder.
3. **Launch:** Open Docker from Applications. Accept the subscription agreement. You may need to grant it system permissions.
4. **Verify Installation:** Open Terminal and run:

```
docker --version
```

6.1 Verify Docker is Working Correctly

Mac users do **not** need to add themselves to the `docker` group — Docker Desktop handles permissions automatically. Run these checks instead:

Check the Docker daemon is running:

```
docker info
```

Should print server details without errors. If you see **Cannot connect to the Docker daemon**, open the Docker Desktop app from Applications and wait for the whale icon in the menu bar to stop animating.

Run the hello-world container:

```
docker run hello-world
```

Expected output ends with **Hello from Docker!** — this confirms Docker can pull and run images.

Check Docker Compose:

```
docker compose version
```

Should print a version like **Docker Compose version v2.29.7**.

List running containers:

```
docker ps
```

Should show an empty table (headers only) — that is success.

Check Docker disk usage (useful before pulling large images like GOWA):

```
docker system df
```

Shows space used by images, containers, and volumes.

6.2 Start Docker Desktop Automatically

To avoid manually launching Docker Desktop after every reboot:

- Open **Docker Desktop** → **Settings** → **General**
- Check “**Start Docker Desktop when you sign in to your computer**”

This ensures `docker` commands work immediately after a reboot.

6.3 Mac-Specific Note — Disk Space

Docker Desktop on Mac stores images inside a virtual disk (`Docker.raw`), typically at:

```
~/Library/Containers/com.docker.docker/Data/vms/0/data/Docker.raw
```

The default size is **64 GB**, which is usually sufficient for GOWA but may fill up over time if you accumulate images.

Check Docker disk usage inside the VM:

```
docker system df
```

Check the actual file size on the Mac host:

```
du -h ~/Library/Containers/com.docker.docker/Data/vms/0/data/Docker.raw
```

If you need more space, increase the disk size in **Docker Desktop** → **Settings** → **Resources** → **Disk image size**.

7 Install Ollama

1. **Download:** Get the macOS installer (`.dmg`) from <https://ollama.com/download/mac>.
2. **Install:** Double-click the `.dmg`, then drag the **Ollama app** to your **Applications** folder.
3. **Launch:** Open Ollama from Applications. On first launch, it will verify the command-line tool is set up. You may be prompted to allow it to create a symlink in `/usr/local/bin` — approve this.
4. **Verify:** Open a new Terminal window and run:

```
ollama --version
```

7.1 Protect Your Disk Space

Models can be tens of gigabytes. If your main drive is small, set a custom model path **before** downloading models.

1. Open **Terminal**.
2. Run this command (replace the path with your actual external drive location):

```
launchctl setenv OLLAMA_MODELS "/Volumes/External/ollama-models"
```

3. **Restart the Ollama app** for the change to take effect.

Part III

Summary and Next Steps

8 Summary of the Two Docker Gotchas

9 Next Steps for Students

Once all three tools (WSL2 on Windows, Docker on both, Ollama on both) are installed and verified, students can proceed with the **Goose + GOWA** setup guide.

Platform	Gotcha	Fix
Windows (WSL2)	permission denied on docker commands	<code>sudo usermod -aG docker \$USER</code> , then restart WSL with <code>wsl -shutdown</code>
Mac	None — Docker Desktop handles permissions	Verify with <code>docker info</code> and <code>docker run hello-world</code>

Where to Run Commands

- **Windows users:** Run all Goose and GOWA commands inside the **Ubuntu terminal** provided by WSL2.
- **Mac users:** Run all Goose and GOWA commands in the native **Terminal**.

10 Quick Verification Checklist

Before moving on, confirm every box is ticked.

Component	Command	Expected
WSL2 (Win)	<code>wsl -status</code>	Default distro: Ubuntu
Docker Engine	<code>docker -version</code>	Version string
Docker Daemon	<code>docker info</code>	No connection errors
Docker Run	<code>docker run hello-world</code>	“Hello from Docker!”
Docker Compose	<code>docker compose version</code>	Version string
Ollama	<code>ollama -version</code>	Version string
Ollama Pull Test	<code>ollama pull qwen3:1.7b</code>	Download completes

You Are Ready

If all checks above pass, you have a fully prepared environment. Proceed to the **Goose + Ollama + GOWA Setup Guide** to build the agent stack.